

Supplementary Statistical and Computational Diagnostics for Low-Turnover Rebalancing for Sparse Index Tracking

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1 Purpose of this supplement

This supplement records the statistical and computational details that support the posterior-informed decisions in the main paper. The purpose is to provide an auditable trail for four objects used repeatedly in the main text: the generalised-Bayes posterior, the empirical-Bayes calibration of the weighted-Laplace sparsity rate, the proximal sampling schemes used to compute posterior summaries, and the diagnostics used to assess whether the reported posterior probabilities are numerically reliable enough for decision support.

The main manuscript reports a conservative sparse-index-tracking workflow. A sparse tracker is constructed once, monitored over rolling holding windows, and locally repaired only when realised tracking deterioration and posterior directional evidence jointly support intervention. The posterior probabilities used in those support and rebalancing rules are computed from proximal Markov chain Monte Carlo (MCMC) output.

Throughout this supplement, $y \in \mathbb{R}^T$ denotes benchmark returns on a fitting window, $R \in \mathbb{R}^{T \times p}$ denotes asset returns, and centred quantities are denoted by y_c and R_c . The same notation is used for construction and for rebalancing, with the change variable denoted by $u = \Delta w$ when appropriate.

Reproducibility: The processed return matrix, ticker universe, rolling-window definitions, implementation notebooks, and CSV outputs used to generate the reported tables and figures are available in the accompanying [GitHub repository](#). The archived CSV outputs include the hold-level performance summaries, rebalancing-decision audit tables, selected (Δw) -grid rows, MCMC diagnostic summaries, portfolio-composition summaries, and the complete (Δw) -grid rows for all holding transitions.

2 Generalised-Bayes decision distributions

2.1 Construction-stage posterior

On a construction fitting window, we use a generalised-Bayes distribution of the form

$$\pi_{\omega, \theta}(w \mid y, R) \propto \exp \left\{ -\omega \ell(w; y, R) - \theta \sum_{j=1}^p \alpha_j |w_j| - \Lambda(\mathbf{1}^\top w - 1)^2 \right\}, \quad (1)$$

where

$$\ell(w; y, R) = \frac{1}{2} \|y_c - R_c w\|_2^2. \quad (2)$$

The learning rate ω is an inverse loss scale. Equivalently, we often write $\omega = 1/\sigma_{\text{eff}}^2$ and minimise

$$\frac{1}{2\sigma_{\text{eff}}^2} \|y_c - R_c w\|_2^2 + \Lambda(\mathbf{1}^\top w - 1)^2 + \theta \sum_{j=1}^p \alpha_j |w_j|. \quad (3)$$

The scale σ_{eff}^2 is not interpreted as a physical residual variance. It is a calibrated loss scale, or temperature, that determines how strongly the tracking loss competes with the sparsity and soft-budget terms. Conditional on each candidate loss scale, the weighted-Laplace rate θ is estimated by stochastic approximation proximal gradient (SAPG), and the resulting MAP/posterior pair is passed to the support-decision layer.

The selected row $c^* = 25$ is the highest-scoring construction row. It yields an effective support of 152, close to the target $n_{\text{FIT}}^* = 150$, while remaining inside the deliberately loose fitting-window

Table S1: Construction-stage effective loss-scale grid for the initial tracker on the first fitting window: 2020/01/03–2021/12/27. The baseline scale is $\hat{\sigma}_{\text{OLS}}^2 = 4.157 \times 10^{-7}$, the reference tracking error is $\text{TE}_{\text{ref}} = 2.926$ bp, and the target effective support is $n_{\text{FIT}}^* = 150$. The selected row is highlighted in bold.

c	$\sigma_{\text{eff}}^2(c)$	$\theta_{\text{EB}}(c)$	$\lambda_{\text{eff}}(c)$	TE_{FIT}	$\text{TE}_{\text{FIT}}/\text{TE}_{\text{ref}}$	n_{raw}	n_{eff}	$\mathbf{1}^\top w$	$d_{\text{FIT}}(c)$
1	4.157×10^{-7}	3.575×10^2	1.486×10^{-4}	5.209	1.780	290	232	0.999037	0.005
10	4.157×10^{-6}	3.436×10^2	1.429×10^{-3}	6.071	2.074	205	188	0.999268	0.315
25	1.039×10^{-5}	3.617×10^2	3.759×10^{-3}	8.134	2.779	170	152	0.999272	0.997
50	2.079×10^{-5}	3.444×10^2	7.159×10^{-3}	10.649	3.639	155	144	0.999310	0.972
75	3.118×10^{-5}	3.531×10^2	1.101×10^{-2}	13.933	4.761	113	103	0.999361	0.171
100	4.157×10^{-5}	3.496×10^2	1.453×10^{-2}	15.133	5.171	157	137	0.999381	0.000
125	5.196×10^{-5}	3.631×10^2	1.887×10^{-2}	17.340	5.926	158	131	0.999313	0.000
150	6.236×10^{-5}	3.523×10^2	2.197×10^{-2}	18.423	6.296	141	116	0.999294	0.000
200	8.314×10^{-5}	3.651×10^2	3.035×10^{-2}	22.177	7.578	131	120	0.999272	0.000

Notes. Tracking errors are reported in basis points, i.e. 10^4 times the return-unit RMSE. The effective MAP shrinkage coefficient is $\lambda_{\text{eff}}(c) = \sigma_{\text{eff}}^2(c) \theta_{\text{EB}}(c)$, which is the relevant scale for comparing MAP penalties across the effective-variance grid. The effective support n_{eff} counts coordinates satisfying $|w_{\text{MAP},j}| \geq 10^{-4}$, whereas n_{raw} counts nonzero coordinates before this numerical threshold. The score $d_{\text{FIT}}(c)$ is the product of the TE feasibility gate and the Gaussian support-size preference.

TE feasibility band. The neighbouring $c = 50$ row has a similar score, but the selected row is closer to the target effective support and is therefore used as the construction-stage posterior for the subsequent support-selection step. We also record the effective MAP penalty

$$\lambda_{\text{eff}}(c) = \sigma_{\text{eff}}^2(c) \theta_{\text{EB}}(c), \quad (4)$$

because this quantity captures the operative shrinkage strength after loss-scale calibration.

Table S2: Initial tracker and one-off floor audit for the final sequential run.

Quantity	Value
Cache label	(150, 25)
Initial support before floor	59
Initial FIT TE target	10.278 bp
Floor threshold	0.0050
Names pruned by floor	6
Mass pruned	0.018060
Support after floor	53
Minimum positive weight after floor	0.005132
One-way floor turnover	1.806%
Recovery target/cap	1.0% / 2.0%

Notes. The one-off floor is part of the implementability variant applied before the sequential holding experiment begins. It removes very small positions from the initially selected 59-name tracker and is distinct from the later recovery-mode repairs.

2.2 Rebalancing posterior on self-financing changes

At a rebalance date, the state variable is the currently held portfolio w_h^{old} . The unknown is not a fresh portfolio, but a self-financing change vector

$$w_h^{\text{new}} = w_h^{\text{old}} + u, \quad \mathbf{1}^\top u = 0. \quad (5)$$

Let

$$y_{c,h}^{\text{res}} = y_{c,h} - R_{c,h} w_h^{\text{old}}. \quad (6)$$

The rebalancing posterior is built on the sum-zero subspace

$$H = \{u \in \mathbb{R}^p : \mathbf{1}^\top u = 0\}, \quad (7)$$

where

$$u \in H, \quad \pi_{\omega_\Delta, \kappa}(u \mid y_h, R_h, w_h^{\text{old}}) \propto \exp \left\{ -\frac{1}{2\sigma_{\Delta, h}^2} \|y_{c, h}^{\text{res}} - R_{c, h} u\|_2^2 - \kappa \sum_{j=1}^p \alpha_j^{(\Delta)} |u_j| \right\}. \quad (8)$$

Here $\sigma_{\Delta, h}^2$ is again an effective rebalancing loss scale, not a literal noise variance. The posterior is used to rank, and gate local trade proposals; final execution is then handled by the bounded local implementation rule in the main paper, which enforces long-only feasibility and turnover limits.

Table S3: Representative Δw effective loss-scale grid before H13. The decision fitting window is 2023-01-03–2024-12-27. The currently held tracker has fitting-window TE 19.024 bp. The selected row is highlighted in bold.

c	$\sigma_\Delta^2(c)$	$\kappa_{\text{EB}}(c)$	TE_{MAP} (bp)	$I_h(c)$ (%)	$n_{\Delta, \text{eff}}$	$\ \Delta w_{\text{MAP}}\ _1$	w_{TE}	w_{ℓ_1}	$d_{\Delta, h}(c)$
1	1.0×10^{-4}	1.194×10^3	17.901	5.9	123	3.596×10^{-2}	1.000	0.957	0.957
10	1.0×10^{-3}	1.194×10^3	17.901	5.9	124	3.597×10^{-2}	1.000	0.956	0.956
25	2.5×10^{-3}	1.194×10^3	17.901	5.9	124	3.598×10^{-2}	1.000	0.956	0.956
50	5.0×10^{-3}	1.194×10^3	17.901	5.9	124	3.598×10^{-2}	1.000	0.956	0.956
75	7.5×10^{-3}	1.194×10^3	17.901	5.9	124	3.598×10^{-2}	1.000	0.956	0.956
100	1.0×10^{-2}	1.194×10^3	17.901	5.9	124	3.598×10^{-2}	1.000	0.956	0.956
125	1.2×10^{-2}	1.194×10^3	17.901	5.9	124	3.598×10^{-2}	1.000	0.956	0.956
150	1.5×10^{-2}	1.194×10^3	17.901	5.9	124	3.598×10^{-2}	1.000	0.956	0.956
200	2.0×10^{-2}	1.194×10^3	17.901	5.9	124	3.598×10^{-2}	1.000	0.956	0.956

Notes. The table reports the full Δw grid for the H13 recovery-mode decision. $I_h(c)$ is the relative fitting-window TE improvement of the MAP proposal over the currently held tracker, expressed as a percentage. The selected row, $c^* = 1$, has $\sigma_\Delta^2 = 10^{-4}$, $\kappa_{\text{EB}} = 1.194 \times 10^3$, $\text{TE}_{\text{MAP}} = 17.901$ bp, $n_{\Delta, \text{eff}} = 123$, and $\|\Delta w_{\text{MAP}}\|_1 = 3.596 \times 10^{-2}$. Complete Δw -grid rows for all holding transitions are provided in the accompanying CSV archive; the supplementary material reports only the selected rows and this representative full grid to keep the presentation readable.

Table S4: Selected rebalancing-grid rows before each post-initial holding window.

Enter hold	Mode	c^*	TE old (bp)	TE MAP (bp)	ΔTE (%)	$n_{\Delta,\text{eff}}$	$\ \Delta w_{\text{MAP}}\ _1$	Implementation
H2	rec.	25	11.820	11.616	1.70	2	8.503e-03	empty
H3	calm	50	12.204	11.952	2.10	3	1.115e-02	empty
H4	calm	25	12.369	12.151	1.80	3	1.033e-02	empty
H5	rec.	1	13.665	13.489	1.30	4	9.556e-03	empty
H6	calm	10	14.071	13.874	1.40	6	1.001e-02	empty
H7	calm	10	14.479	14.277	1.40	5	1.018e-02	empty
H8	calm	1	15.065	14.878	1.20	5	1.022e-02	empty
H9	calm	1	15.300	15.134	1.10	5	9.778e-03	empty
H10	rec.	1	15.682	15.494	1.20	7	1.096e-02	sell-only/no dest.
H11	rec.	1	17.598	17.281	1.80	26	1.668e-02	buy-dom. top-up
H12	rec.	75	19.098	18.494	3.20	65	2.418e-02	empty
H13	rec.	1	19.024	17.901	5.90	123	3.596e-02	buy-dom. top-up
H14	rec.	1	21.248	19.298	9.20	204	5.138e-02	buy-dom. top-up
H15	rec.	1	21.169	19.483	8.00	158	3.981e-02	buy-dom. top-up
H16	rec.	1	21.492	19.770	8.00	164	4.078e-02	buy-dom. top-up

Notes. The MAP proposal is used for posterior exploration and candidate generation. The final implemented decision may be no trade if the posterior directional gates or local funding rules do not approve a feasible repair. Complete Δw -grid rows for all holding transitions are provided in the CSV archive.

3 Empirical-Bayes calibration by SAPG

3.1 Construction-stage update

For a fixed loss scale, the weighted-Laplace prior has unnormalised density

$$\exp\{-\theta g(w)\}, \quad g(w) = \sum_{j=1}^p \alpha_j |w_j|. \quad (9)$$

Because g is one-homogeneous, the derivative of the log-prior normalising constant contributes p/θ to the marginal-likelihood score. It is convenient to update $\eta = \log \theta$. The SAPG recursion used in the implementation is

$$\eta_{k+1} = \Pi_{[\log \theta_{\min}, \log \theta_{\max}]} \left\{ \eta_k + \rho_k \left(p - \theta_k g(w^{(k)}) \right) \right\}, \quad \theta_k = \exp(\eta_k), \quad (10)$$

where $w^{(k)}$ is obtained from a MYULA transition targeting the current smoothed posterior, and ρ_k is a Robbins–Monro step size. The empirical-Bayes estimate is the Polyak–Ruppert average on the log scale,

$$\bar{\eta} = \frac{\sum_{k > k_{\text{burn}}} \omega_k \eta_k}{\sum_{k > k_{\text{burn}}} \omega_k}, \quad \theta_{\text{EB}} = \exp(\bar{\eta}). \quad (11)$$

The initialisation used here is a scale-matching rule. It is not a second tuning rule; it only gives SAPG a numerically stable starting range. The calculation is recorded next, since it fixes the initial box used by the construction-stage SAPG recursion.

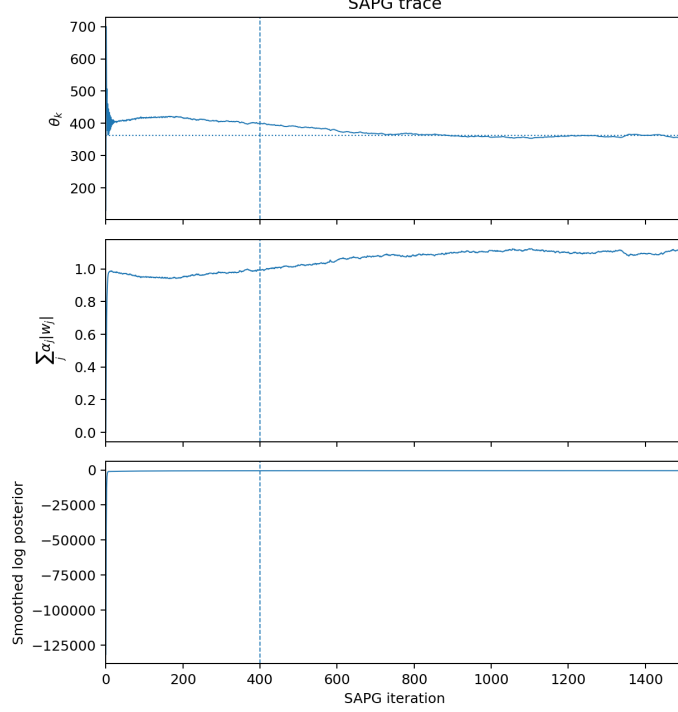


Figure S1: Representative SAPG and posterior trace plots from the construction stage.

3.2 Initial scale heuristic for θ_0

This subsection records the calculation that justifies the heuristic (“method-of-moments”) choice of θ_0 . On the FIT window, we compute an initial least-squares solution

$$w'_{\text{LS}} = R_c^\dagger y_c, \quad (12)$$

which we project¹ to the affine hyperplane

$$\mathcal{C} = \{w : \mathbf{1}^\top w = 1\} \quad (13)$$

to obtain

$$w_{\text{LS}} = w'_{\text{LS}} - \frac{\mathbf{1}^\top w'_{\text{LS}} - 1}{p} \mathbf{1}, \quad (14)$$

which satisfies the budget constraint.

Consider a weighted Laplace prior with independent coordinates,

$$\pi(w \mid \theta) \propto \prod_{j=1}^p \frac{\theta \alpha_j}{2} \exp\{-\theta \alpha_j |w_j|\}, \quad (15)$$

and coordinate precision $\lambda_j = \theta \alpha_j$. For a scalar Laplace random variable $W \sim \text{Laplace}(0, 1/\lambda)$,

$$\mathbb{E}|W| = \frac{1}{\lambda}, \quad \mathbb{E}\{\lambda|W|\} = 1. \quad (16)$$

¹The Euclidean projector has the form $P_C(w) = w - \{(\mathbf{1}^\top w - 1)/p\} \mathbf{1}$.

Thus, under the weighted independent Laplace reference law,

$$\mathbb{E}\{\theta\alpha_j|w_j|\} = 1, \quad \mathbb{E}\left\{\theta\sum_{j=1}^p\alpha_j|w_j|\right\} \approx p. \quad (17)$$

Informally, a typical prior draw satisfies

$$\theta\sum_{j=1}^p\alpha_j|w_j| \approx p. \quad (18)$$

We choose θ_0 so that the reference vector w_{LS} has approximately this prior penalty scale:

$$\theta_0\sum_{j=1}^p\alpha_j|(w_{\text{LS}})_j| \approx p, \quad (19)$$

which yields

$$\theta_0 \approx \frac{p}{\sum_{j=1}^p\alpha_j|(w_{\text{LS}})_j|}. \quad (20)$$

Because the weights α_j are normalised to have mean approximately one, this may be read as an average-scale matching condition: on average, $\alpha_j|(w_{\text{LS}})_j|$ is matched to the prior mean absolute scale $1/\theta_0$. Equivalently, the heuristic chooses about one unit of weighted Laplace penalty per coordinate.

In practice, we regularise the initial value as

$$\theta_0 = \max\left\{\frac{p}{\sum_{j=1}^p\alpha_j|(w_{\text{LS}})_j|}, 10^{-6}\right\}, \quad (21)$$

and constrain the SAPG updates to the box

$$\Theta = [\theta_0/10, 10\theta_0]. \quad (22)$$

The lower bound prevents θ_0 from becoming numerically very small when the least-squares reference is noisy or poorly identified. The one-order-of-magnitude box allows the SAPG recursion to adapt the scale parameter while preventing drift to extremely small values, which would make the prior nearly flat, or extremely large values, which would over-shrink the posterior geometry.

3.3 Δw update on the sum-zero subspace

For the rebalancing posterior, the prior acts on $u = \Delta w$ constrained to the $(p-1)$ -dimensional subspace H . Let

$$S_{\Delta}(u) = \sum_{j=1}^p\alpha_j^{(\Delta)}|u_j|. \quad (23)$$

The corresponding log-scale SAPG update is

$$\zeta_{k+1} = \Pi_{[\log \kappa_{\min}, \log \kappa_{\max}]} \left\{ \zeta_k + \rho_k \left(d - \kappa_k S_{\Delta}(u^{(k)}) \right) \right\}, \quad \kappa_k = \exp(\zeta_k), \quad (24)$$

with $d = p-1$. The dimension correction changes from p to $p-1$ because the prior is restricted to the self-financing hyperplane.

4 Proximal posterior sampling

4.1 Moreau–Yosida smoothing and proximal maps

Let

$$G_\theta(w) = \theta \sum_{j=1}^p \alpha_j |w_j|. \quad (25)$$

The implementation smooths the scaled nonsmooth term G_θ , not merely the unscaled ℓ_1 norm. The Moreau–Yosida envelope is

$$G_{\theta,\lambda}(w) = \min_{u \in \mathbb{R}^p} \left\{ G_\theta(u) + \frac{1}{2\lambda} \|u - w\|_2^2 \right\}, \quad (26)$$

and

$$\nabla G_{\theta,\lambda}(w) = \frac{1}{\lambda} \left(w - \text{prox}_{\lambda G_\theta}(w) \right). \quad (27)$$

For the construction-stage weighted ℓ_1 term, the proximal map is component-wise soft-thresholding,

$$\left[\text{prox}_{\lambda G_\theta}(w) \right]_j = \text{sign}(w_j) \max\{|w_j| - \lambda \theta \alpha_j, 0\}. \quad (28)$$

For the rebalancing problem, the nonsmooth term also includes the sum-zero constraint. The proximal map is therefore

$$\text{prox}_{\lambda(\kappa S_\Delta + \iota_H)}(z) = \arg \min_{u: \mathbf{1}^\top u = 0} \left\{ \frac{1}{2} \|u - z\|_2^2 + \lambda \kappa \sum_{j=1}^p \alpha_j^{(\Delta)} |u_j| \right\}. \quad (29)$$

Our implementation computes this as weighted soft-thresholding with a scalar shift/Lagrange multiplier chosen so that the output sums to zero. This is the important computational distinction between the construction sampler and the Δw sampler.

4.2 MYULA and preconditioned MALA

For a smoothed potential Φ_λ , MYULA iterates

$$x^{(k+1)} = x^{(k)} - \delta \nabla \Phi_\lambda(x^{(k)}) + \sqrt{2\delta} \xi^{(k)}, \quad \xi^{(k)} \sim N(0, I_p). \quad (30)$$

MYULA is used inside SAPG and as a cheap exploratory sampler. For final posterior summaries, the main runs use a preconditioned MALA proposal

$$x' = x - \delta P^2 \nabla \Phi_\lambda(x) + \sqrt{2\delta} P \xi, \quad \xi \sim N(0, I_p), \quad (31)$$

followed by the usual Metropolis–Hastings correction for the smoothed target.

For construction-stage weight sampling, the diagonal preconditioner is based on the Jacobi curvature approximation

$$P = D^{-1/2}, \quad D = \text{diag} \left\{ \frac{1}{\sigma_{\text{eff}}^2} \text{diag}(R_c^\top R_c) + 2\Lambda \mathbf{1} \right\}. \quad (32)$$

For the Δw sampler, the self-financing condition is handled through the proximal map, so the diagonal preconditioner is based on the data-Hessian part of the current fitting window.

The Metropolis–Hastings correction uses the Gaussian forward and backward proposal densities. If

$$q(x' | x) = N\left(x - \delta P^2 \nabla \Phi_\lambda(x), 2\delta P^2\right), \quad (33)$$

then

$$\log a(x, x') = -\Phi_\lambda(x') + \Phi_\lambda(x) + \log q(x | x') - \log q(x' | x), \quad (34)$$

and the proposal is accepted with probability $\min\{1, \exp(\log a(x, x'))\}$.

5 Posterior sampling diagnostics

5.1 Autocorrelation, effective sample size, and Monte Carlo error

Let $z^{(1)}, \dots, z^{(M)}$ be a scalar time series extracted from post-burn MCMC output. The empirical autocorrelation at lag k is denoted by $\hat{\rho}_k$. The integrated autocorrelation time is estimated by a truncated sum,

$$\hat{\tau} = 1 + 2 \sum_{k=1}^K \hat{\rho}_k, \quad (35)$$

where K is chosen by a standard truncation rule, for example, the first nonpositive autocorrelation or a fixed maximum lag. The effective sample size is

$$\text{ESS}(z) = \frac{M}{\hat{\tau}}. \quad (36)$$

For each posterior draw $w^{(m)}$, define the realised fitting-window tracking error

$$\text{TE}^{(m)} = \left\{ \frac{1}{T} \sum_{t=1}^T \left(y_t - r_t^\top w^{(m)} \right)^2 \right\}^{1/2}, \quad (37)$$

using the same realised-return convention as in the main paper. Then $\text{ESS}(\text{TE})$ is computed by applying the scalar ESS formula to $\{\text{TE}^{(m)}\}_{m=1}^M$. If $\widehat{\text{sd}}(\text{TE})$ is the empirical standard deviation of these values, the Monte Carlo standard error of the posterior mean TE is approximated by

$$\text{MCSE}(\overline{\text{TE}}) \approx \frac{\widehat{\text{sd}}(\text{TE})}{\sqrt{\text{ESS}(\text{TE})}}. \quad (38)$$

For support and trade decisions, scalar TE diagnostics are not enough. We also report ESS values for decision-relevant coordinates. For the construction run, these are large MAP coordinates, selected-support coordinates, and borderline support coordinates. For rebalancing, they are candidate Δw_j coordinates appearing in the raw recovery set or near the directional-probability threshold.

5.2 Construction-stage diagnostics

The construction-stage posterior is used to estimate practical-positive probabilities of the form

$$\hat{P}(w_j > \varepsilon_0 \mid y, R) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\{w_j^{(m)} > \varepsilon_0\}. \quad (39)$$

Table S5 therefore summarises mixing both for scalar tracking diagnostics and for support-decision coordinates.

Table S5: Construction-stage posterior sampling diagnostics for the selected tracker posterior. Tracking-error diagnostics use the fitting window and are reported in basis points.

Quantity	Value
Post-burn draws	390,000
Acceptance rate	0.549
Tuned γ	4.841e-03
TE posterior mean (bp)	25.958
TE posterior sd (bp)	1.031
ESS(TE)	1003.9
MCSE(TE mean, bp)	0.0325
Min ESS, selected support	973.5
Min ESS, borderline selected	973.8
Min ESS, borderline rejected top-50	973.9
Wall time (s)	1525.9

5.3 Rebalancing-stage diagnostics

The rebalancing posterior is used to estimate directional probabilities

$$\hat{p}_{\text{dir},h,j}(\varepsilon) = \begin{cases} M^{-1} \sum_{m=1}^M \mathbf{1}\{\Delta w_{h,j}^{(m)} > \varepsilon\}, & \Delta w_{\text{MAP},h,j} > 0, \\ M^{-1} \sum_{m=1}^M \mathbf{1}\{\Delta w_{h,j}^{(m)} < -\varepsilon\}, & \Delta w_{\text{MAP},h,j} < 0. \end{cases} \quad (40)$$

Accordingly, Table S6 focuses on recovery-date candidate coordinates rather than on all p coordinates.

Table S6: Posterior sampling diagnostics for the sequential Δw MALA runs.

Enter hold	γ	Accept	TE mean (bp)	TE 90% interval (bp)	ESS(TE)	min ESS $_{\Delta}$	Wall (s)
H2	5.96e-04	0.631	29.83	[25.00, 37.17]	10422	6880	223.6
H3	8.39e-04	0.554	44.81	[37.42, 55.37]	12190	2090	203.3
H4	7.99e-04	0.570	32.13	[27.09, 39.32]	11255	5628	208.3
H5	7.51e-04	0.598	14.75	[13.96, 15.70]	7317	3928	870.0
H6	7.25e-04	0.595	22.35	[19.63, 25.89]	9912	5678	235.6
H7	7.31e-04	0.588	22.63	[19.94, 26.16]	8838	5702	270.6
H8	7.25e-04	0.612	15.95	[15.17, 16.84]	7292	3648	879.4
H9	7.43e-04	0.606	16.22	[15.45, 17.08]	7816	9726	894.3
H10	7.55e-04	0.617	16.61	[15.82, 17.49]	7771	7730	897.0
H11	8.45e-04	0.625	18.52	[17.56, 19.56]	6474	6323	891.1
H12	9.34e-04	0.606	61.38	[53.10, 71.84]	11596	3741	213.8
H13	1.15e-03	0.599	19.51	[18.05, 21.18]	5192	9940	879.4
H14	1.24e-03	0.607	20.97	[19.23, 22.96]	5568	12005	870.9
H15	9.47e-04	0.594	20.66	[18.99, 22.56]	6558	7527	881.7
H16	9.85e-04	0.578	20.97	[19.27, 22.93]	5565	9790	881.6

Each rebalancing posterior uses 250,000 proposals with thinning by 6, retaining 36,666 post-burn-in samples after discarding 5,000 kept draws. The final column reports wall-clock time for the long run only. The TE summaries in this table are posterior diagnostics on the decision fitting-window Δw posterior samples; they are not forecasts of the next holding-window TE and are not the realised TE of the implemented held portfolio.

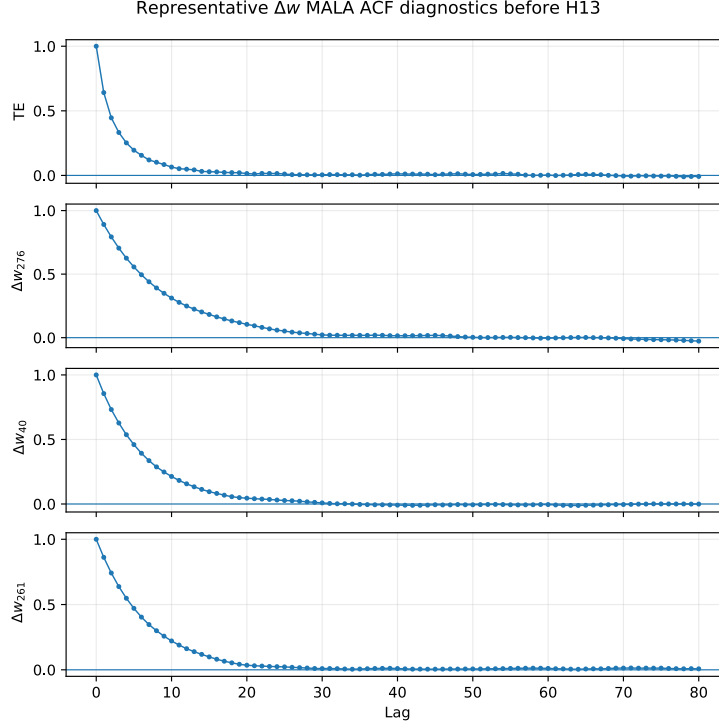


Figure S2: Representative MALA autocorrelation diagnostics for the rebalancing posterior before H13. The panels report autocorrelation functions for the tracking-error functional and for three selected Δw coordinates: the largest approved buy-side move (276), a large approved MAP move (40), and a near-threshold MAP move (261) that was not implemented. The rapid decay towards zero indicates satisfactory mixing for both the scalar tracking diagnostic and the inspected coordinate-level movements.

6 Posterior decision audit

6.1 Construction support sensitivity

The main paper uses a practical-positive support rule

$$S(\varepsilon_0, \pi_+^*) = \left\{ j : w_{\text{MAP},j} > \varepsilon_0, \hat{P}(w_j > \varepsilon_0 \mid y, R) \geq \pi_+^* \right\}. \quad (41)$$

The purpose of the supplementary sensitivity table is not to re-optimize the support, but to show that the selected operating point is not an isolated artefact of one threshold pair.

Table S7: Practical-positive support sensitivity for the construction-stage posterior. The selected case-study rule is highlighted in bold.

ε_0	π_+^*	$ S(\varepsilon_0, \pi_+^*) $	MAP mass	mean $\hat{P}_+$	min. $\hat{P}_+$	top-50 MAP	comment
1.0×10^{-3}	0.60	59	0.582	0.831	0.610	29	selected
1.0×10^{-3}	0.65	56	0.563	0.842	0.652	28	
1.0×10^{-3}	0.70	45	0.495	0.883	0.702	24	
1.0×10^{-3}	0.80	33	0.402	0.925	0.804	19	
2.5×10^{-3}	0.60	48	0.560	0.813	0.606	29	
2.5×10^{-3}	0.65	43	0.534	0.836	0.656	28	
2.5×10^{-3}	0.70	35	0.458	0.874	0.721	22	
2.5×10^{-3}	0.80	25	0.384	0.922	0.840	18	
5.0×10^{-3}	0.60	32	0.483	0.812	0.614	26	
5.0×10^{-3}	0.65	26	0.424	0.855	0.663	22	
5.0×10^{-3}	0.70	24	0.413	0.869	0.702	22	
5.0×10^{-3}	0.80	20	0.367	0.897	0.807	18	
1.0×10^{-2}	0.60	12	0.319	0.801	0.610	12	
1.0×10^{-2}	0.65	10	0.293	0.838	0.733	10	
1.0×10^{-2}	0.70	10	0.293	0.838	0.733	10	
1.0×10^{-2}	0.80	4	0.168	0.940	0.857	4	

Notes. The table varies the practical holding threshold ε_0 and posterior probability threshold π_+^* in the construction-stage rule

$$S(\varepsilon_0, \pi_+^*) = \left\{ j : w_{\text{MAP},j} > \varepsilon_0, \hat{\text{Pr}}(w_j > \varepsilon_0 \mid y, R) \geq \pi_+^* \right\}.$$

“MAP mass” is the fraction of MAP absolute weight retained by the selected support, $\hat{P}_+$ denotes the empirical posterior practical-positive probability over the selected names, and “top-50 MAP” counts how many selected names belong to the 50 largest MAP coordinates by absolute weight. The selected rule $(\varepsilon_0, \pi_+^*) = (10^{-3}, 0.60)$ yields 59 names before the subsequent one-off implementability floor.

6.2 Recovery candidate ranking and implementation

For recovery-mode decisions, candidate moves first pass a practical MAP-size gate,

$$|\Delta w_{\text{MAP},h,j}| \geq \varepsilon_\Delta, \quad \varepsilon_\Delta = 10^{-4}.$$

Directional support is then measured by the posterior sign probability

$$\hat{p}_{\text{dir},h,j}(0) = \begin{cases} M^{-1} \sum_{m=1}^M \mathbf{1}\{\Delta w_{h,j}^{(m)} > 0\}, & \Delta w_{\text{MAP},h,j} > 0, \\ M^{-1} \sum_{m=1}^M \mathbf{1}\{\Delta w_{h,j}^{(m)} < 0\}, & \Delta w_{\text{MAP},h,j} < 0. \end{cases}$$

Eligible recovery moves must satisfy

$$\hat{p}_{\text{dir},h,j}(0) \geq p_{h,j}^{\text{req}},$$

and are ranked by

$$r_{h,j} = \hat{p}_{\text{dir},h,j}(0) |\Delta w_{\text{MAP},h,j}|.$$

Here $\hat{p}_{\text{dir},h,j}(0)$ is the posterior sign probability in the MAP direction, while ε_Δ enters separately through the practical MAP-size gate $|\Delta w_{\text{MAP},h,j}| \geq \varepsilon_\Delta$. The score is deliberately simple: it prevents the recovery budget from being allocated only to high-probability but economically tiny moves, or only to large MAP moves with weak directional support.

Table S8: Sequential decision audit for the final tracker.

Hold	$ S $	Case	Turnover (%)	Trades	Hold TE (bp)	No-trade reason
H1		initial	0.000	0	17.151	
H2	0	empty	0.000	0	15.363	empty candidate set
H3	0	empty	0.000	0	12.218	empty candidate set
H4	0	empty	0.000	0	19.064	empty candidate set
H5	0	empty	0.000	0	14.599	empty candidate set
H6	0	empty	0.000	0	13.119	empty candidate set
H7	0	empty	0.000	0	14.755	empty candidate set
H8	0	empty	0.000	0	14.776	empty candidate set
H9	0	empty	0.000	0	20.460	empty candidate set
H10	1	sell-only/no dest.	0.000	0	26.887	sell-only; no approved destination
H11	5	buy-dom. top-up	1.000	7	27.634	
H12	0	empty	0.000	0	19.754	empty candidate set
H13	10	buy-dom. top-up	1.000	12	36.074	
H14	10	buy-dom. top-up	1.000	12	23.183	
H15	10	buy-dom. top-up	1.000	12	24.274	
H16	10	buy-dom. top-up	1.000	13	29.582	

The table reports the implemented decision entering each hold. H10 is a useful no-trade diagnostic: posterior-approved moves were sell-only, so the method did not create turnover without an approved destination for the released budget.

6.3 No-trade diagnostics

No-trade decisions are part of the method, not missing actions. We explicitly document the key cases where the posterior or implementation rule suppresses trading. In the final branch, H10 is the canonical sell-only/no-destination diagnostic, while H12 demonstrates that the algorithm does not continue trading mechanically after the H11 repair.

Table S9: No-trade diagnostics for recovery-mode decisions.

Diagnostic item	H10	H12
Recovery status	Triggered by the realised-TE monitor.	Triggered by the realised-TE monitor.
Final candidate rule	Recovery-mode asymmetric sign gate $p_{dir}(0)$, with practical MAP-size gate $\varepsilon_{\Delta} = 10^{-4}$.	Recovery-mode asymmetric sign gate $p_{dir}(0)$, with practical MAP-size gate $\varepsilon_{\Delta} = 10^{-4}$.
Approved set	$ S = 1$: one sell and no approved buys.	$ S = 0$: no coordinate passes the recovery gate.
Posterior evidence	The approved move is a GOOG sell with $p_{dir} = 0.535$; no posterior-approved destination buy is available.	No move passes the final recovery gate, although the MAP proposal would reduce the fitting-window TE.
Decision-FIT TE	Old tracker: 15.682 bp; MAP proposal: 15.494 bp; implemented tracker: 15.682 bp.	Old tracker: 19.098 bp; MAP proposal: 18.494 bp; implemented tracker: 19.098 bp.
Implemented action	No rebalance; zero turnover and zero trades.	No rebalance; zero turnover and zero trades.
Realised holding-window TE	26.887 bp.	19.754 bp, down from 27.634 bp in H11 after the previous 1% repair.
Interpretation	The rule suppresses sell-only actions because reducing exposure without an approved destination would create turnover without a self-financing posterior-supported repair.	The portfolio is held unchanged after H11; the method does not continue trading mechanically once the recovery gate no longer approves a candidate set.

Notes. The table reports final implemented decisions, not just raw Δw MAP proposals. H10 is the canonical sell-only/no-destination diagnostic. H12 is included because it shows that the maintenance layer can return to inaction after a previous recovery repair.

7 Internal tracker comparisons

This section reports internal comparisons used to assess the robustness and operating point of the proposed maintenance workflow. First, we compare three initial tracker constructions labelled by (n_{target}, c^*) , where n_{target} is the construction-stage target effective support and c^* is the selected loss-scale multiplier. This comparison motivates the (150, 25) branch used in the main case study, without claiming that it is universally optimal. Second, for the selected (150, 25) branch, we compare the MH-corrected MALA sampler used in the main paper with an inexact preconditioned MYULA alternative. Third, we examine more active recovery rules to illustrate how the same posterior-informed maintenance architecture can move along the tracking-error–turnover frontier when a larger repair budget is acceptable.

7.1 Variants for the base tracker

This subsection summarises the internal branch comparison for three initial trackers labelled by the pair (n_{target}, c^*) . The first component is the target effective support used in the initial construction-stage grid score, and the second component is the selected loss-scale multiplier. The three branches are therefore denoted (180, 10), (150, 25), and (130, 50).

All three runs use the same sequential maintenance rule: a one-off 0.5% implementability floor is applied before the first holding window; the conservative recovery rule then uses asymmetric posterior-directional gates for existing positions, new buys, and sells; approved recovery trades are ranked by posterior-weighted MAP magnitude; and implemented recovery trades target 1% one-way

turnover subject to a 2% hard cap and local funding. Thus, the comparison isolates the effect of the initial construction branch, conditional on the same downstream maintenance rule.

Table S10: Full-horizon comparison of the three floor-adjusted internal branches.

Branch (n_{target}, c^*)	Initial names	Mean TE	Median TE	Max TE	Max hold	H1–H8 mean TE	H9–H16 mean TE	Turn. (%)	Trades	Final names
(180,10)	61	22.341	21.823	41.909	13	15.038	29.645	4.05	62	63
(150,25)	53	20.556	19.409	36.074	13	15.131	25.981	5.00	56	55
(130,50)	48	24.031	20.011	39.933	13	16.906	31.155	2.00	16	50

Notes: TE values are in basis points. ‘Initial names’ is the number of nonzero holdings after the one-off 0.5% floor and before the first holding-window evaluation. The branch labels refer to the construction-stage target effective support and selected loss-scale multiplier, not to the post-floor number of holdings.

Full-horizon summary. The (150, 25) branch is the best-performing internal branch in this set. It has the lowest full-horizon mean TE, median TE, and maximum TE, despite using a middle-sized implemented support after the floor. The denser (180, 10) branch has the best early-window average, but it deteriorates more strongly in the late windows and has the largest maximum TE. The sparser (130, 50) branch trades the least, but has the weakest late-horizon tracking performance. This supports using (150, 25) as the main internal branch: it occupies the most favourable compromise among the three tested construction choices under the same conservative maintenance rule.

Table S11: Per-holding-window realised TE for the three internal branches.

Hold	(180, 10)	(150, 25)	(130, 50)
1	14.305	17.151	17.507
2	14.398	15.363	19.477
3	12.844	12.218	14.182
4	21.780	19.064	20.545
5	15.976	14.599	16.606
6	12.230	13.119	13.584
7	13.825	14.755	17.913
8	14.943	14.776	15.436
9	21.866	20.460	18.710
10	28.021	26.887	29.777
11	33.986	27.634	34.427
12	23.659	19.754	24.398
13	41.909	36.074	39.933
14	26.472	23.183	35.221
15	29.190	24.274	27.412
16	32.056	29.582	39.363

Per-window tracking errors. The three branches behave similarly in the first half of the experiment, with all average realised TEs between roughly 15 and 17 bp over holding windows 1–8. The main separation occurs after holding window 9. The (150, 25) branch remains the most stable late-horizon branch, while both (180, 10) and (130, 50) experience larger deterioration in the

shock/recovery portion of the sample. All branches have their maximum TE in holding window 13, confirming that this window is the common stress point rather than an artefact of a single construction choice.

Table S12: Implemented rebalancing activity under the conservative recovery rule.

Branch	Trading windows	Total turn. (%)	Total trades	Max one-window turn. (%)
(180,10)	11, 13, 14, 15, 16	4.047	62	1.000
(150,25)	11, 13, 14, 15, 16	5.000	56	1.000
(130,50)	12, 13	2.000	16	1.000

Rebalancing activity. The recovery layer remains conservative across all three branches. The (130, 50) branch implements only two recovery trades, for total one-way turnover of 2%. The (150, 25) branch spends 5% total one-way turnover across the full horizon, while the denser (180, 10) branch spends about 4.05%. These turnover levels are far below those of rolling full-reconstruction baselines and are consistent with the intended interpretation of the proposed method as a maintenance layer rather than a repeated reconstruction engine.

Discussion. The internal branch comparison gives two useful messages for the empirical case study. First, the branch selected for the main paper is not an arbitrary single run: among the three floor-adjusted construction choices examined here, (150, 25) gives the best realised tracking behaviour under the same sequential maintenance rule. Second, the comparison illustrates the role of the construction-stage loss scale. A lower loss-scale multiplier and larger target effective support do not automatically improve realised tracking after posterior support selection, the one-off floor, and conservative rebalancing. Conversely, the more aggressively shrunk (130, 50) branch is cheaper to maintain but gives up too much late-horizon tracking performance. The (150, 25) branch is therefore a reasonable main branch for the subsequent comparison with reconstruction baselines.

7.2 Inexact MCMC: MALA versus pMYULA

The main case study uses a Metropolis–Hastings correction for the long MALA runs. This correction removes the proposal-discretisation bias relative to the smoothed Moreau–Yosida target. The remaining smoothing approximation is part of the proximal-MCMC construction. Here, we examine the performance of the inexact preconditioned MYULA sampler, which omits the MH correction.

Our results below show that the full H1–H16 sampler comparison is almost unchanged at the decision level. Replacing the MH-corrected sequential sampler by preconditioned MYULA without MH correction changes the full-horizon mean tracking error by only 0.017 bp and leaves the final number of names unchanged. The no-MH branch makes one additional tiny H10 trade of 0.014% one-way turnover; otherwise the recovery pattern and late-horizon behaviour are essentially the same.

Table S13: Full-horizon comparison of the MH-corrected and no-MH sequential samplers.

Metric	MALA	pMYULA
Mean TE (bp)	20.56	20.57
Median TE (bp)	19.41	19.50
Max TE (bp)	36.07	35.85
Mean TE (H9–H16) (bp)	25.98	26.02
Median TE (H9–H16) (bp)	25.58	25.59
Holds > 25 bp	4	4
Turnover (%)	5.00	5.01
Trade windows	5	6
Final names	55	55
Top-25 mass (%)	72.8	73.0

Table S14: Per-hold late-horizon TE, one-way turnover, and trade counts. Turnover and trades are executed before evaluating the corresponding holding window.

Hold	MALA			pMYULA		
	TE (bp)	TO (%)	Trades	TE (bp)	TO (%)	Trades
10	26.89	0.00	0	26.85	0.01	3
11	27.63	1.00	7	27.97	1.00	8
12	19.75	0.00	0	19.94	0.00	0
13	36.07	1.00	12	35.85	1.00	10
14	23.18	1.00	12	23.34	1.00	12
15	24.27	1.00	12	24.34	1.00	12
16	29.58	1.00	13	29.39	1.00	13

Discussion. The no-MH chain targets a discretised and smoothed approximation. Nevertheless, relative to the MALA version, the downstream decision layer—MAP direction, marginal directional probabilities, thresholding, truncation, local funding, and bounded implementation—is quantitatively very close in this case study. We therefore retain the MH-corrected MALA sampler as the main specification, while reporting pMYULA as a computational sensitivity: it is substantially faster, but its approximation bias must be regarded as part of the numerical trade-off.

7.3 Sensitivity to more active recovery rules

The main case study deliberately reports a conservative recovery rule: it trades rarely, spends only a small pre-specified turnover budget, and avoids additional trading-rule complexity.

Here we explore more active recovery variants. The purpose is to show that the same posterior-informed maintenance engine can accommodate different operating points along the TE–turnover frontier. Variant A increases the recovery budget, allows more approved moves, and permits sell-only recovery cases to be paired with weak posterior-supported destination buys. Variant B adds a fit-window escalation trigger and a weak fallback gate when the strict recovery set is empty. These variants improve late-horizon TE in this sample, but they do so at the expected cost of higher turnover, greater rule complexity, and a more concentrated final portfolio.

The late-horizon experiments begin at H9 because the main branch makes no trades before that point; the tracker entering H9 is therefore the post-floor 53-name portfolio from the full sequential run. The H7 and H8 realised tracking errors are used to initialise the recovery trigger, so the H9–H16

runs use the same recovery-state information that would have been available in the full sequential experiment. Calm-mode rules are unchanged. Only recovery-mode rules are relaxed.

The main (conservative) branch uses the asymmetric recovery rule from the main workflow: existing buys/sells require $p_{\text{dir}} \geq 0.525$, new buys require $p_{\text{dir}} \geq 0.575$, the approved list is truncated to at most 10 moves and 4 new buy names, the target one-way recovery turnover is 1%, and the hard cap is 2%.

Variant A is the simpler active-recovery specification. It allows sell-only recovery to choose weak posterior-supported destination buys, increases the approved list to 25 moves and 8 new buy names, targets 2.5% one-way turnover, and uses a 5% hard cap. In other words, Variant A spends a larger recovery budget and allows posterior-supported destinations for sell-only cases. Variant B adds a fit-window deterioration trigger and a weak fallback gate when the strict recovery set is empty.

Table S15: Recovery-rule specifications compared in this section. All branches use the same cached (150,25) tracker after the one-off 0.5% implementability floor.

Branch	Recovery rule	Interpretation
Main Tracker	Existing buys/sells require $p_{\text{dir}} \geq 0.525$; new buys require $p_{\text{dir}} \geq 0.575$; sell-only recovery is not executed without a posterior-approved destination; at most 10 moves and 4 new buys; target one-way turnover 1%, hard cap 2%.	Conservative main branch. The rule spends little turnover and treats recovery as a bounded local repair.
Variant A	Recovery-only active variant: sell-only recovery may be funded into weak posterior-supported destination buys; at most 25 moves and 8 new buys; target one-way turnover 2.5%, hard cap 5%; no fit-window fallback.	Simpler active-recovery specification: spend more local turnover after deterioration while keeping the posterior-directional gates and no global renormalisation.
Variant B	Variant A plus a fit-window deterioration trigger, $\text{TE}_{\text{old}}^{\text{fit}} \geq 17$ bp. If the strict recovery set is empty when the fit trigger fires, use a weak fallback gate: existing buys/sells $p_{\text{dir}} \geq 0.50$ and new buys $p_{\text{dir}} \geq 0.525$.	More active recovery specification. It tests whether a fit-window warning can reduce the late shock before it appears in the realised holding-window TE.

Table S16: Late-horizon H9–H16 comparison between the conservative branch and two active recovery variants.

Metric	Main Tracker	Variant A	Variant B
Mean TE (bp)	25.98	21.79	21.45
Median TE (bp)	25.58	20.71	21.27
Max TE (bp)	36.07	32.69	26.73
Holds > 25 bp	4	1	1
Turnover (%)	5.00	10.00	15.00
Trade windows	5	4	6
Max hold turnover (%)	1.00	2.50	2.50
Final names	55	44	38
Top-25 mass (%)	72.8	77.0	83.4

Table S17: Per-hold late-horizon TE, one-way turnover, and trade intensity. Turnover and trades are executed before evaluating the corresponding holding window.

Hold	Main Tracker			Variant A			Variant B		
	TE (bp)	TO (%)	Trades	TE (bp)	TO (%)	Trades	TE (bp)	TO (%)	Trades
10	26.89	0.00	0	20.01	2.50	7	20.01	2.50	7
11	27.63	1.00	7	21.83	0.00	0	23.98	2.50	16
12	19.75	0.00	0	17.17	0.00	0	15.30	2.50	16
13	36.07	1.00	12	32.69	0.00	0	26.73	0.00	0
14	23.18	1.00	12	20.97	2.50	16	24.62	2.50	11
15	24.27	1.00	12	19.03	2.50	13	18.44	2.50	8
16	29.58	1.00	13	22.14	2.50	12	22.07	2.50	15

Discussion. The per-hold results explain why the active variants help but do not completely eliminate the late shock. In Variant A, the method trades at H10 and then does not trade at H11–H13 because the strict posterior recovery candidate set is empty. H13 therefore remains the single stress hold, with TE about 32.69 bp. Variant B uses the fit-window escalation and weak fallback at H11 and H12, spending 2.5% one-way turnover in each of those decisions; this reduces the subsequent H13 TE to about 26.73 bp. However, the rule still does not push H13 below 25 bp, and it does so at the cost of materially higher turnover and concentration.

The conservative main branch already gives the intended low-turnover behaviour: over H9–H16 it spends only 5% one-way turnover, while recovering from the late shock through bounded local repairs. Variant A shows that the engine can spend more turnover when the investor wants a more active response: mean late-horizon TE falls from 25.98 bp to 21.79 bp, and the number of late holds above 25 bp falls from four to one. The price is a doubling of late-horizon turnover and a more concentrated final portfolio. Variant B further reduces the maximum late-horizon TE, from 32.69 bp in Variant A to 26.73 bp, but requires 15% late-horizon turnover and ends with only 38 active names.